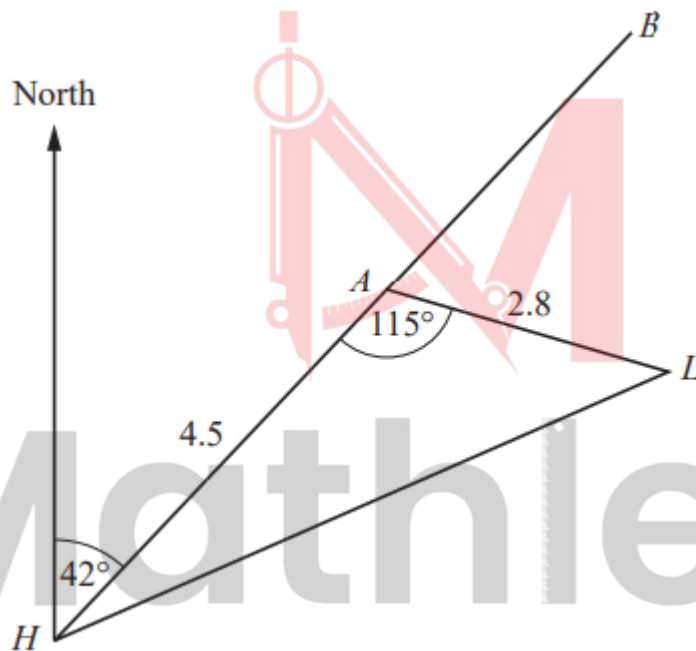


Trigonometry

Worksheet 3a

Bearing (Non-Calculator)

1. S08/qp2/q9(a)



The diagram shows the positions of a harbour, H, a lighthouse, L, and two buoys A and B.

HAB is a straight line.

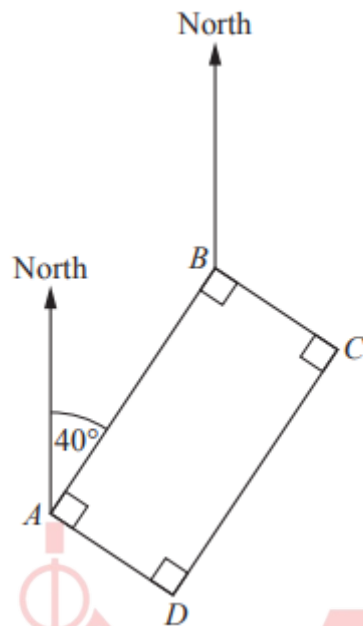
The bearing of A from H is 042° .

$HA = 4.5$ km, $AL = 2.8$ km and $\angle HAL = 115^\circ$.

Find the bearing of

- i. H from A. [1]
- ii. L from A. [1]

2. S15/qp11/q20(a)

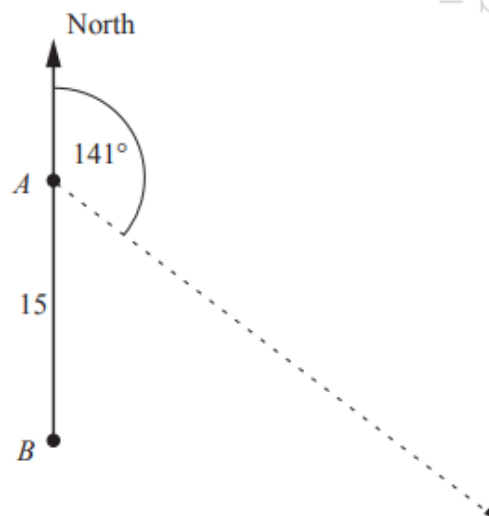


Four oil-rigs are positioned at the vertices of a rectangle ABCD . The bearing of B from A is 040°. Find the bearing of

- | | | |
|------|-----------|-----|
| i. | A from B | [1] |
| ii. | C from B, | [1] |
| iii. | C from D. | [1] |

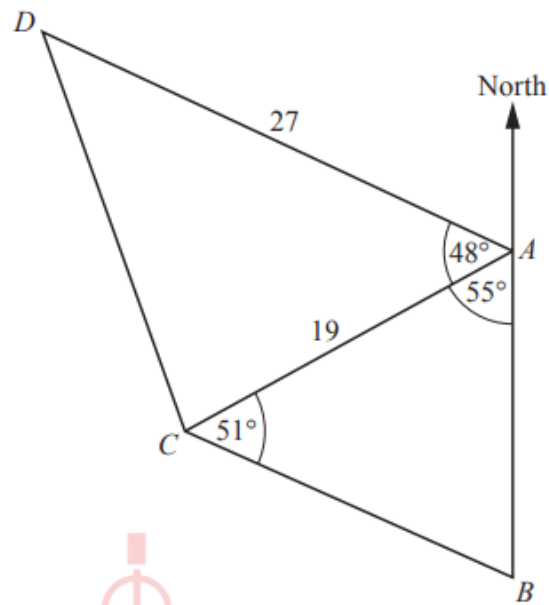
3. S15/qp22/q8(a)

Two ports, A and B, are 15km apart and B is due south of A. A boat sails from A on a bearing of 141°.



State the bearing of A from the boat. [1]

4. W16/qp21/q7(a,b)

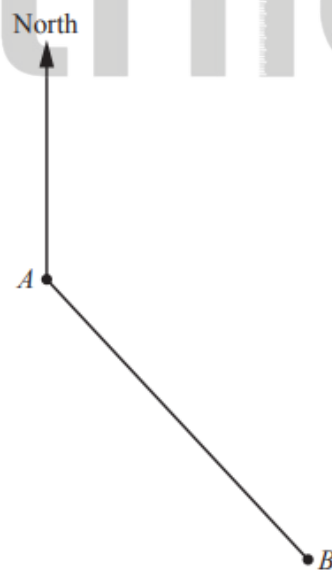


The diagram shows the positions of four islands at A, B, C and D. A is due north of B. $\angle DAC = 48^\circ$, $\angle CAB = 55^\circ$ and $\angle BCA = 51^\circ$. $AC = 19\text{km}$ and $AD = 27\text{km}$.

- Calculate the bearing of D from A. [1]
- Calculate the bearing of A from C. [1]

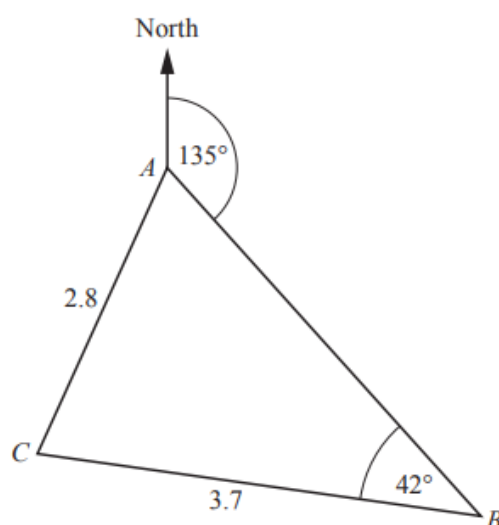
5. S17/qp12/q5

The diagram shows the position of two villages A and B.



- Measure the bearing of B from A. [1]
- The bearing of village C from A is 265° . Work out the bearing of A from C. [1]

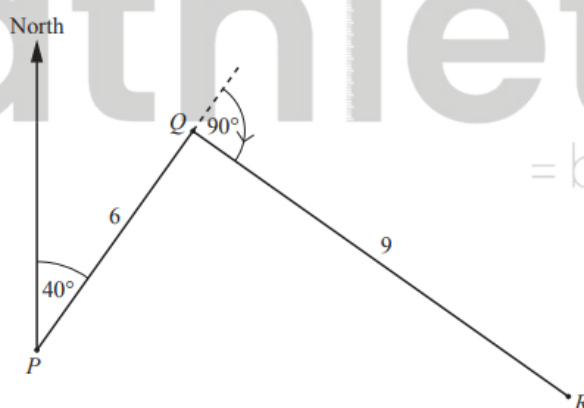
6. S18/qp22/q7 (Use calculator here)



A yacht sails the triangular route shown. The bearing of B from A is 135° . $BC = 3.7\text{km}$, $AC = 2.8\text{km}$ and $\hat{ABC} = 42^\circ$.

- Show that $\hat{CAB} = 62.2^\circ$, correct to 1 decimal place. [3]
- Find the bearing of A from C. [2]
- The yacht sails from A to B to C to A. Calculate the total length of the route. [4]

7. W18/qp11/q16(b)

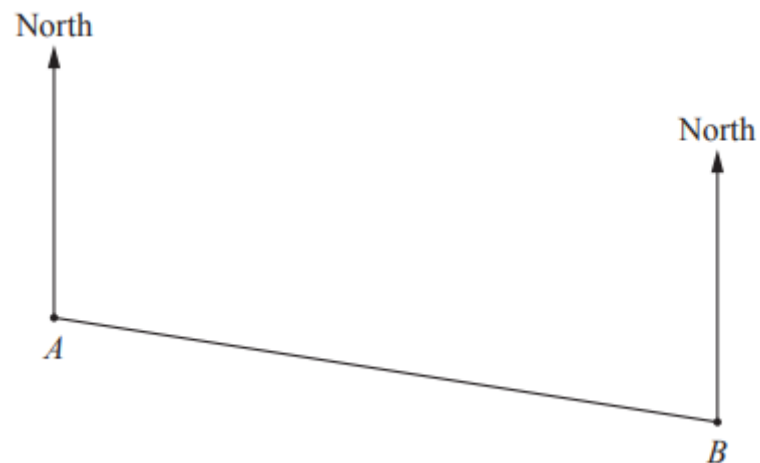


A boat travels from P to Q. At Q, it turns through 90° and travels to R as shown in the diagram. It then returns from R to Q, and then to P, following the same route in reverse. $PQ = 6\text{ km}$ and $QR = 9\text{ km}$. The first part of the journey, from P to Q to R, takes 3 hours. The return part of the journey, from R to Q to P, takes 2 hours.

The bearing of Q from P is 040° .

- Calculate the bearing of R from Q. [1]
- Calculate the bearing of P from Q. [1]

8. S20/qp22/q9 (0580)

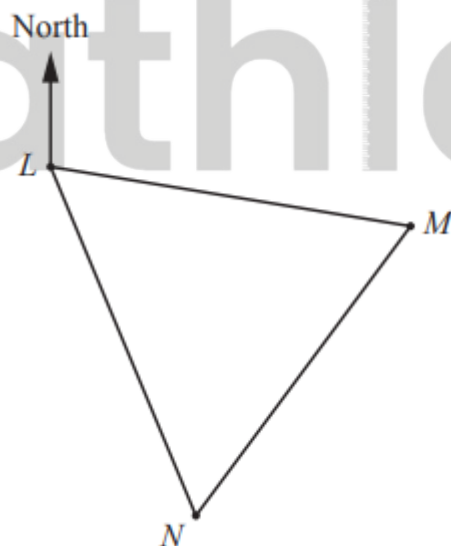


The bearing of B from A is 105° . Find the bearing of A from B. [2]

9. S20/qp23/q8 (0580)

The bearing of X from Y is 274° . Calculate the bearing of Y from X. [2]

10. M21/qp22/q10 (0580)



On a map, the positions of the towns L, M and N form an equilateral triangle. The bearing of M from L is 103° . Work out the bearing of L from N. [2]

Answer Key

1. S08/qp2/q9(a)

9	(a)	(i)	222° 107°	B1 B1	[2]
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2. S15/qp11/q20(a)

20	(a)	(i)	220°	1	
		(ii)	130°	1	

3. S15/qp22/q8(a)

8	(a)	321	1	
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4. W16/qp21/q7(a,b)

7	(a)	283°	1	
	(b)	055°	1	

5. S17/qp12/q5

5(a)	137	1	
5(b)	085	1	

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6. S18/qp22/q7

7(a)	$\sin CAB = \frac{3.7 \sin 42}{2.8}$ OR $\hat{CAB} = \sin^{-1}\left(\frac{3.7 \sin 42}{2.8}\right)$ OR $\frac{\sin CAB}{3.7} = \frac{\sin 42}{2.8} \text{ and }$ $\sin = 0.88[42\dots]$	M2	M1 for $\frac{3.7}{\sin CAB} = \frac{2.8}{\sin 42}$ oe
	$\hat{CAB} = 62.15[4\dots]$	A1	
7(b)	[0]17.2°	2	M1 for $135 + 62.2 - 180$ oe
7(c)	10.5 to 10.6	4	B3 for 4.05 to 4.06 OR M2 for $\sqrt{2.8^2 + 3.7^2 - 2 \times 2.8 \times 3.7 \times \cos(180 - 42 - 62.2)}$ oe or M1 for $2.8^2 + 3.7^2 - 2 \times 2.8 \times 3.7 \times \cos(180 - 42 - 62.2)$ oe OR M2 for $\frac{2.8 \sin(180 - 42 - 62.2)}{\sin 42}$ oe or M1 for $\frac{\sin(180 - 42 - 62.2)}{AB} = \frac{\sin 42}{2.8}$ oe OR M2 for $\frac{3.7 \sin(180 - 42 - 62.2)}{\sin 62.2}$ oe or M1 for $\frac{\sin(180 - 42 - 62.2)}{AB} = \frac{\sin 62.2}{3.7}$ oe OR B1 for $\hat{ACB} = 75.8$

7. W18/qp11/q16(b)

16(b)(i)	130°	1	
16(b)(ii)	220°	1	

8. S20/qp22/q9 (0580)

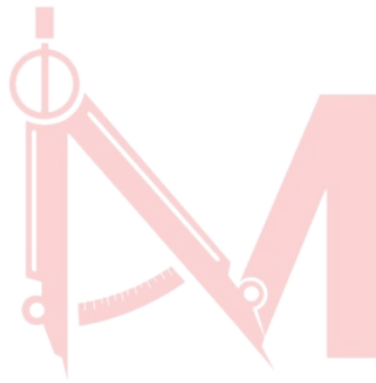
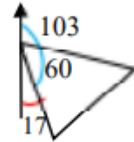
9	285	2	M1 for $180 + 105$ or 75 or 105 seen in correct position at B
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9. S20/qp23/q8 (0580)

8	[0]94	2	M1 for 86 or $274 - 180$ or for sketch with 274 marked correctly
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10. M21/qp22/q10 (0580)

10	343	2	B1 for 103 in correct position and 60 or 17 in correct position
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